INCH-POUND

MIL-PRF-1/290E
22 Februalry 2008
SUPERSEDING
MIL-E-1/290D
1 February 1971

PERFORMANCE SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING

TYPE OA2WA

Inactive for new design
after 7 March 1997

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the electron tube described herein shall consist of this document and MIL-PRF-1.

DESCRIPTION: Voltage regulator, miniature.

Outline - - - 5-3 (EIA).

Base - - - E7-1.

Envelope - - - T5-1/2.

Cathode - - - Glow discharge.

Base connections:

Pin No.	- - -	1	2	3	4	5	6	7
Element	- - -	a	k	int con	k	a	int con	k

ABSOLUTE RATINGS:

Parameter:	Total darkness ionization voltage	Ambient light ionization voltage	Operating voltage	Operating current	TA	TE	Alt
Unit:	V dc	V dc	V dc	mA dc	°C	°C	ft
Maximum:	165	165	158	30	---	150	Note 7
Minimum:	---	---	140	5	-55	---	---
Test conditions:	---	---	---	---	---	---	---

GENERAL:

Qualification - Required.

Reliable tube.

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TABLE I. Testing and Inspection.

MIL-STD-1311 method	Requirement or test	Conditions	Acceptance level (see note 10)	Symbol	Limits		Unit
					Min	Max	
	Qualification inspection						
1031	Variable-frequency vibration	Rp = 10,000 ohms; Ebb/Ib = 20 mA dc	---	Ep	---	100	mV ac
	Conformance inspection, part 1 (see note 1)						
3347	Ionization voltage (1)	Ebb/Ib = 5-30 mA dc; illumination = 5 to 50 ft candles (see note 2)	0.4	Ez	---	165	V dc
3337	Voltage drop (1)	Ebb/Ib = 30 mA dc	0.4	Etd	144	153	V dc
3337	Voltage drop (2)	Ebb/Ib = 5 mA dc	0.4	Etd	144	153	V dc
3335	Regulation	(1) Etd minus (2) Etd	0.4	Reg	---	±5	V dc
1201	Short and discontinuity detection		0.4	---	---	---	---
	Conformance inspection, part 2 (see note 1)						
3278	Noise	Ebb/Ib = 30 mA dc	1.0	Eb	---	5	mV ac
3278	Oscillation	Esig = 100 mV ac; Ebb/Ib = 5-30 mA dc	1.0	---	---	---	---
---	Voltage jump	Ebb/Ib - 5-30 mA dc (see note 3)	2.5	Jump	---	600	mV dc
3347	Ionization voltage (2)	See note 4	2.5	Ez	---	165	V dc
3305	Leakage	Eb = 50 V dc; Rp = 3,000 ohms	2.5	Llb	---	5	µA dc
3337	Voltage drop (3)	Ebb/Ib = 20 mA dc	2.5	Etd	144	153	V dc
---	Repeatability	Ebb/Ib = 10 mA dc (see note 5)	2.5	Etd	---	600	mV dc
1031	Low-frequency vibration	Rp = 10,000 ohms; Ebb/Ib = 20 mA dc	2.5	Ep	---	100	mV ac
1041	Shock	900 G	---	---	---	---	---
1031	Vibration-fatigue test	2.5 G; fixed frequency; F = 25 min, 60 max	2.5	---	---	---	---

See notes at end of table.

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TABLE I. Testing and Inspection. -Continued.

MIL-STD-1311 method	Requirement or test	Conditions	Acceptance level (see note 10)	Symbol	Limits Min	Limits Max	Unit
	Conformance inspection, part 2 (see note 1) -Continued						
---	Post-shock and vibration-fatigue test end points	Low-frequency vibration	---	Ep	---	100	mV ac
		Ionization voltage (1)	---	Ez	---	165	V dc
		Voltage drop (1)	---	Etd	142	155	V dc
		Voltage drop (2)	---	Etd	142	155	V dc
		Regulation	---	Reg	---	±5	V dc
1121	Base strain		---	---	---	---	---
2126	Glass strain		2.5	---	---	---	---
1105	Permanence of marking		---	---	---	---	---

MIL-STD-1311 method	Requirement or test	Conditions	Symbol	Limits Min	Limits Max	Unit
	Conformance inspection, part 3					
1516	Stability life test	Ebb/Ib = 20 mA dc; TA = room	---	---	---	---
---	Stability life-test end point	Change in voltage drop (3) of individual tubes (see note 9)	ΔEtd t	---	2.0	V dc
1521	Survival-rate life-test	Stability life-test, or equivalent conditions	---	---	---	---
---	Survival-rate life-test end points	Short and discontinuity detection.	---	---	---	---
		Change in voltage drop (3) of individual tubes.	ΔEtd t	---	5.0	V dc
---	Life test	Stability life-test or equivalent conditions, TE = 150°C (min) (see note 8)	---	---	---	---
---	Life-test end points (500 hours)	Regulation	Reg	---	±6	V dc
		Voltage drop (1)	Etd	142	155	V dc
		Voltage drop (2)	Etd	142	155	V dc
		Voltage drop (3)	Etd	142	155	V dc
		Change in voltage drop (3) of individual tubes	Etd	---	6	V dc
		Ionization voltage (1)	Ez	---	165	V dc

See notes at end of table.

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TABLE I. Testing and Inspection. -Continued.

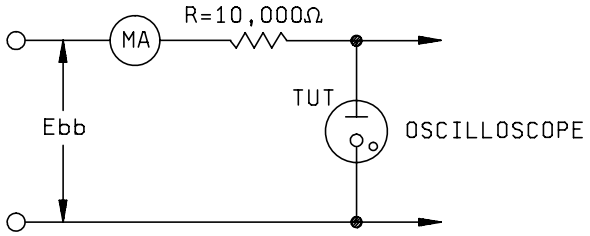
MIL-STD-1311 method	Requirement or test	Conditions	Symbol	Limits		Unit
				Min	Max	
	Conformance inspection, part 3 - Continued					
---	Life-test end points (1,000 hours)	Regulation	Reg	---	±8	V dc
		Voltage drop (1)	Etd	140	158	V dc
		Voltage drop (2)	Etd	140	158	V dc
		Voltage drop (3)	Etd	140	158	V dc
		Change in voltage drop (3) of individual tubes	Etd	---	8	V dc
		Ionization voltage (1)	Ez	---	165	V dc

NOTES:

1. Sampling shall be in accordance with accept on zero defect sampling plan in MIL-PRF-1, table III, using the acceptance level tabulated above.

2. This test to be performed at the conclusion of the holding period.

3. Vary current from 5 mA dc to 30 mA dc and back to 5 mA dc (by adjusting Ebb slowly). Sudden jumps registered on the oscilloscope shall be not greater than the specified value.



4. Conditions for this test shall be those of ionization voltage (1), except testing shall be done in total darkness and the tube shall not have conducted or been exposed to light for at least 24 hours prior to testing. The tube shall fire within 20 seconds maximum.

5. The tube shall be tested in the following manner:

   a. The voltage drop shall be read at 10 mA dc drain.

   b. The tube shall be turned off for 1 minute.

   c. The tube shall be restarted and operated at the same current.

   d. Etd shall be read after 1 minute of operation.

   e. The "on-off" cycle shall be repeated a minimum of five times. The maximum difference in tube voltage drop shall be taken as the measure of repeatability.

6. This test shall be conducted on the initial lot and thereafter on a lot approximately every 6 months. When one lot has passed, the 6-month rule shall apply. In the event of lot failure, the lot shall be rejected and the succeeding lots shall be subjected to this test until a lot passes.

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TABLE I. Testing and Inspection. -Continued.

7. See "Reduced pressure (altitude) rating", and altitude, maximum peak voltage.

8. Envelope temperature (TE) requirements, when measured in accordance with the temperature by conduction-band measurement (MIL-STD-1311, method 1226), will be satisfied if a tube having bogey tube drop (ETD ± 3 percent) under normal test conditions, is determined to operate at or above minimum specified temperature at any position in the life-test rack.

9. Duration shall be one hour.

10. This specification sheet utilizes an accept on zero defect sampling plan in accordance with MIL-PRF-1, table III.

11. Revision letters are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Referenced documents. In addition to MIL-PRF-1, this specification sheet references MIL-STD-1311.

Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army - CR
Navy - EC
Air Force - 11
DLA - CC

Preparing activity:
DLA - CC

(Project 5960-2008-015)

Review activities:
Army - AR, MI
Navy - AS, CG, MC
Air Force - 99

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

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